

EXAMINATION ASSIGNMENT

AI Virtual Try-On Chrome Extension

Digital Body Profile | Product Detection | AI Try-On | Cross-Website Experience

1. Objective

Build a Chrome extension that allows a user to create a personal digital profile and virtually try products from other websites on themselves. The extension should identify products displayed on a shopping website, allow the user to select a product, and generate a realistic visualization of the user wearing or using it.

2. Core Concept

The user first creates a personal digital profile using their own photographs. Once the profile is created, the user can browse an external shopping website. When a product is visible on the page, the Chrome extension should allow the user to select that product and choose 'Try On'.

The basic workflow is:

1. Create personal digital profile.
2. Open any supported shopping website.
3. Click the Chrome extension.
4. Extension identifies available products on the page.
5. User selects a product.
6. User clicks 'Try On'.
7. Product image and user profile are sent to the AI try-on system.
8. System generates a realistic visualization.
9. Result is displayed to the user.

3. Personal Digital Profile

The user should be able to create a reusable digital profile that can be used for future try-on requests.

The profile should support appropriate photographs such as:

- Front/full-body photograph
- Upper-body photograph
- Leg/body photograph
- Foot/shoe photograph
- Face photograph
- Additional photographs where required by the AI model

The system should guide the user on how photographs should be captured for the best results. The student may design the profile format and determine what images are required for different product categories.

4. Product Categories

The system should be designed to support multiple product types, including:

- T-shirts and tops
- Shirts
- Dresses
- Jackets
- Pants and trousers
- Shoes
- Jewellery
- Necklaces
- Accessories

The system architecture should allow additional categories to be added later without rebuilding the complete extension.

5. Chrome Extension

The primary user interface must be a Chrome extension. It should work while the user is browsing an external website rather than requiring the user to manually download and upload every product image.

- Extension popup or side panel.
- Detect product images or product information on the current page.
- Display selectable products.
- Provide a clear 'Try On' action.
- Show processing status.
- Display the generated result.
- Handle pages with multiple products.

6. Product Detection

The extension must investigate the current webpage and identify the product being viewed or products available on the page.

The system may use available information such as:

- Product images
- Image URLs
- Product title
- Product description
- Structured product data
- Page metadata
- Product variants
- Visible price/product information

The extension should work across multiple website structures rather than being hard-coded for only one website.

7. AI Virtual Try-On

The central challenge is generating a realistic result using the user's digital profile and the selected product.

Examples:

- A T-shirt should appear naturally on the user's upper body.
- A dress should appear naturally on the user's body with appropriate fit and positioning.
- Shoes should appear on the user's feet with appropriate perspective.
- A necklace should appear around the user's neck.
- Other accessories should be positioned appropriately.

The generated result should preserve important characteristics of the user while realistically incorporating the selected product.

8. Context-Aware Visualization

The system should not simply paste a product image over a person. Where appropriate, the generated image should have a suitable environment, pose and presentation.

For example, a beach-style T-shirt could be visualized in a suitable beach or outdoor environment while maintaining the user's identity and the product's appearance.

9. Product Accuracy

The selected product should remain recognizable and visually consistent with the source product.

- Colors should be preserved as accurately as practical.
- Patterns and prints should be preserved.
- Logos and important product details should not be unnecessarily altered.
- Product proportions should be appropriate.
- The result should match the selected product rather than generating a generic replacement.

10. Multiple Product Images

When a product has multiple images, the system should determine which image or combination of images provides the best input for the try-on process.

Where useful, the user should be able to choose a product image or variant before starting the try-on.

11. User Experience

The experience should be simple enough for a normal shopper to understand without technical knowledge.

10. Create Profile
11. Browse Shopping Website
12. Open Extension
13. Select Product
14. Try On
15. Wait for Generation

16. View Result

12. Supported Website Testing

Students must demonstrate the extension on multiple real shopping websites. The extension should not be built only around one fixed website.

Students should select websites with different page structures and product categories to demonstrate that their product-detection approach is reusable.

13. Backend & AI Architecture

The extension may communicate with a backend service that manages profile data, product information, AI model requests and generated results.

Students must document:

- Chrome extension architecture
- Product detection mechanism
- Profile storage
- Backend/API architecture
- AI model/provider used
- Image-processing workflow
- Result storage
- Error handling

14. Privacy & Security

Personal photographs are sensitive user data and must be handled carefully.

- Do not expose user profile images publicly.
- Use secure communication between the extension and backend.
- Do not expose API keys inside the Chrome extension.
- Clearly identify where profile images are stored.
- Allow the user to delete their profile and generated results.
- Do not use personal images for model training unless the user has explicitly provided appropriate permission.

15. Performance

The try-on experience should be designed to minimize unnecessary waiting.

- Optimize product-image retrieval.
- Avoid repeatedly uploading the same profile images.
- Use appropriate image sizes.
- Show generation progress.
- Handle failed AI requests gracefully.
- Cache reusable information where appropriate.

16. Mandatory Demonstration

- Create a personal digital profile.
- Install and demonstrate the Chrome extension.
- Open multiple shopping websites.
- Detect products from the webpages.
- Select a product from the extension.
- Generate a virtual try-on result.
- Demonstrate at least two different product categories.
- Show that the product remains recognizable in the generated image.
- Demonstrate the profile being reused for multiple products.

17. Required Deliverables

- Chrome extension source code
- Backend source code, if applicable
- Working extension build
- AI try-on integration
- Database/schema, if applicable
- Technical architecture documentation
- Setup and installation instructions
- Demonstration video
- Live or accessible demonstration environment where applicable

18. Bonus Challenges

- Support additional shopping websites automatically.
- Support more product categories.
- Try-on directly from product listing pages containing many products.
- Try multiple products in one generated outfit.
- Side-by-side comparison of products.
- Save previous try-on results.
- Virtual wardrobe.
- Multiple user profiles.
- Improved automatic product-category detection.
- Automatic selection of the best product image.
- Pose-aware try-on.
- More realistic environment/background generation.
- Fast or low-latency generation mode.
- Mobile browser or companion application.

19. Important Constraints

- The extension must be designed for legitimate use on websites and products the user is permitted to interact with.
- Do not bypass website security mechanisms or access restrictions.
- The assignment is focused on the technical challenge of product detection, image understanding, AI generation and user experience.

20. Expected Outcome

The final system should feel like a practical AI shopping assistant. A user creates their digital profile once, visits an online shopping website, selects a product through the Chrome extension, and receives a realistic visualization of how that product could look on them.

Examination Focus: Chrome Extension • Computer Vision • AI Image Generation • Product Detection • UX